

San Francisco State University



**Math 895
Research Project
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Final Project Progress Report
Statistical & Machine Learning Analysis of Airbnb
Listings in NYC (2024)

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⇒ Executive Summary

The growth of short term rental platforms has reshaped the accommodation market in New York City. This project conducts a comprehensive statistical and machine learning analysis of more than twenty thousand Airbnb listings in order to understand how pricing, host strategy, listing characteristics, and guest engagement interact to form the structure of this market. The study integrates descriptive analytics, inferential statistics, supervised learning models, and unsupervised clustering techniques to reveal how listings are priced, how hosts manage their properties, and how guests respond through reviews and engagement.

The dataset used for this analysis includes detailed information on price, room type, availability, host identity, geographic location, review activity, and listing ratings. Rigorous data preparation steps were applied, including consistency checks, cleaning of missing values, outlier treatments, variable transformations, and feature encoding. Exploratory analysis showed clear differences across boroughs and room types, along with notable behavioral patterns among hosts with single listings compared to those managing multiple listings.

The methodological framework for this project combines several complementary approaches. Statistical tests were used to confirm price differences across borough and room type categories. Regression modeling identified the strongest predictors of price, with room type, location, property size, and host type emerging as primary drivers. Additional models were used to analyze guest engagement through review activity. Classification algorithms were trained to distinguish high demand from low demand listings, and their comparison demonstrated that nonlinear ensemble models provide substantially stronger predictive performance than linear baselines. The project concludes with clustering analysis that identifies meaningful market segments based on price, host behavior, reviews, ratings, and availability.

The findings indicate that supply side characteristics are the strongest determinants of price and engagement in the New York City Airbnb market. Location and room type have consistent and substantial effects on nightly price. Host strategy influences both pricing and engagement, although engagement differences diminish once listing availability is taken into account. High demand listings generally combine competitive pricing, positive ratings, active review histories, and lean operational strategies. The clusters produced by the unsupervised analysis reveal several distinct segments that illustrate the diversity of market strategies used by different hosts.

This project provides a data driven view of how Airbnb listings operate within New York City. The insights gained are relevant for hosts seeking to optimize pricing and guest experience, for platform stakeholders evaluating listing performance, and for analysts studying the dynamics of peer to peer accommodation markets.

⇒ Introduction: Background & Problem Statement

Short term rental platforms have become a central part of the hospitality landscape in major cities. New York City represents one of the largest and most active markets for Airbnb, with thousands of listings operating across diverse neighborhoods and host types. The rapid expansion of this market has raised important questions regarding how listings are priced, how host behavior shapes market outcomes, and how guests engage with different types of properties. Understanding these dynamics is essential for hosts who want to optimize their strategies, for platform stakeholders interested in market structure, and for policymakers concerned with the economic and social implications of short term rentals.

This project analyzes more than twenty thousand Airbnb listings in New York City in order to investigate the factors that influence price, demand, and host performance. The dataset contains detailed information including nightly price, room type, geographic location, availability, ratings, review activity, and host characteristics. These variables create a rich multidimensional environment that can be studied using descriptive analytics, statistical inference, predictive modeling, and clustering. Because the dataset captures both supply side attributes, such as room type and host strategy, and demand side signals, such as review behavior, it enables a holistic analysis of the short term rental market.

The analysis begins with a full review of the dataset and the preparation steps required to make the data suitable for statistical and machine learning methods. This includes addressing missing values, correcting structural issues, identifying and treating outliers, transforming skewed variables, and engineering new features such as log transformed price, log transformed engagement metrics, and host type categories. The exploratory analysis then provides an intuitive understanding of how price levels, host behaviors, availability patterns, and review activity differ across boroughs and listing types. Initial insights from this phase guide the selection of appropriate methods for more formal modeling.

The methodological framework for the project involves both statistical testing and predictive modeling. Statistical tests are used to evaluate whether price differences across boroughs and room types are significant, whether host strategies influence pricing, and whether engagement patterns differ between single listing hosts and multi listing hosts. Regression models are used to quantify the effect of listing characteristics on price and on guest engagement, incorporating both linear and nonlinear relationships. Classification models are used to distinguish high demand from low demand listings, and the comparison of linear and ensemble classifiers provides insight into the complexity of the underlying demand structure. Clustering analysis is used to uncover natural market segments based on aggregated listing characteristics including price, host behavior, ratings, and review patterns.

Research Objectives

The analysis conducted in this project is organized around a set of research objectives that aim to capture the core economic and behavioral patterns observed within the New York City Airbnb market. These objectives are to determine how nightly prices vary across boroughs and room types, to identify the listing and location characteristics that most strongly predict price, to examine whether host strategy influences pricing behavior, to evaluate how host strategy relates to guest engagement through review activity, to classify listings into high demand and low demand segments based on review patterns, and to identify natural clusters of listings that emerge when considering price, host attributes, ratings, and reviews. Together, these objectives provide a structured framework for understanding how supply side and demand side factors interact within this market.

⇒ Dataset Description, Cleaning & Preparation

- **Dataset Description**

This study uses a dataset of 20,770 Airbnb listings located within New York City. The dataset was obtained in comma separated values format and contains a range of listing level attributes that describe property characteristics, host behavior, geographic location, availability, review activity, and guest ratings. Each row represents a unique listing entry and the attributes collectively provide a multifaceted view of the supply side and demand side factors that characterize the short term rental market.

The dataset includes structured variables describing each listing. Core property information consists of nightly price, room type classification, number of beds, number of bedrooms, number of bathrooms, and property rating. Host level attributes include the host identification number, host name, and the number of listings managed by that host. Location related variables include the borough identifier, detailed neighborhood name, and the geographic coordinates in terms of latitude and longitude. Indicators of market activity include the number of reviews, reviews per month, and the number of reviews recorded in the last twelve months. Measures of listing availability, such as the number of days the listing is available within a calendar year, provide insight into supply side constraints and booking patterns.

Collectively, the dataset contains both numerical variables and categorical variables. Numerical variables capture monetary measures, counts, availability values, and ratings. Categorical variables represent host types, room types, borough categories, and neighborhood names. The richness of the dataset allows a thorough investigation of pricing patterns, demand indicators, and host strategies. It also provides a suitable foundation for statistical modeling and machine learning tasks aimed at predicting prices, classifying demand levels, and discovering latent clusters in the market.

Before conducting any analysis, it is essential to ensure that the dataset is structurally consistent, free from problematic entries, and suitable for statistical techniques. The following subsections describe the cleaning procedures applied to the dataset, the handling of missing values, the treatment of outliers, and the transformations carried out to prepare the data for regression, classification, and clustering models.

Variable Name	Type	Description
id	Numeric (Identifier)	Unique identifier assigned to each listing. Later removed because of duplication issues.
name	Text	Title of the listing as provided by the host. Not used for modeling.
host_id	Numeric	Unique identifier representing the host associated with the listing.
host_name	Text	Name of the host. Not used for statistical modeling.
neighbourhood_group	Categorical	Borough in which the listing is located. Categories include Manhattan, Brooklyn, Queens, Bronx, and Staten Island.
neighbourhood	Text	Neighborhood name within the borough. Informational only.
latitude	Numeric	Geographic latitude coordinate of the listing.
longitude	Numeric	Geographic longitude coordinate of the listing.
room_type	Categorical	Type of accommodation offered. Categories include Entire home or apartment, Private room, Shared room, and Hotel room.
price	Numeric	Nightly rental price in United States dollars.
minimum_nights	Numeric	Minimum number of nights required per booking.
number_of_reviews	Numeric	Total number of reviews received by the listing.
last_review	Date	Date of the most recent review. Converted to datetime format for consistency.
reviews_per_month	Numeric	Average number of reviews received per month. Key indicator of demand.
availability_365	Numeric	Number of days in a year the listing is available for booking.
calculated_host_listings_count	Numeric	Number of listings managed by the host. Used to classify host types.
number_of_reviews_ltm	Numeric	Number of reviews received in the last twelve months.
license	Categorical	License status of the listing. Not used in predictive models.
rating	Numeric	Guest rating score. Converted to numeric and used in models.
bedrooms	Categorical / Numeric	Number of bedrooms. Recast into numeric form during cleaning.
beds	Numeric	Number of beds offered.

Variable Name	Type	Description
baths	Categorical / Numeric	Number of bathrooms. Converted to numeric form during cleaning.
unique_id	Numeric (Identifier)	Synthetic unique identifier created to replace duplicated id column.
host_type	Categorical	Derived variable identifying the host as Single (one listing) or Multi (multiple listings).
log_price	Numeric (Transformed)	Log transformed nightly price computed as ($\log(1 + \text{price})$).
log10_rpm	Numeric (Transformed)	Log base 10 transformed reviews per month computed as ($\log_{10}(1 + \text{reviews per month})$).
log10_engagement_adj	Numeric (Transformed)	Log base 10 transformed availability adjusted engagement metric.

• Date Cleaning

A structured data cleaning process was applied to ensure that the dataset was analytically reliable and ready for statistical testing and machine learning. The raw data exhibited inconsistencies in format, duplicated identifiers, mixed numeric and text values, and several fields that required standardization. The following steps summarize the cleaning procedures performed on the dataset.

- Correction of inconsistent data types for key variables. The `last_review` field was converted to a standardized datetime format, and values stored as text in the `rating`, `bathrooms`, and `bedrooms` fields were cleaned and converted into numeric form so they could be used in modeling.
- Removal of an unreliable identifier column. The original listing identifier contained thousands of duplicate entries and did not uniquely represent observations. To preserve row level uniqueness, this column was removed and replaced with a synthetic `unique_id`.
- Standardization of categorical fields. Variables such as `room_type` and `neighbourhood_group` were checked for consistency. A new categorical variable named `host_type` was created to distinguish between hosts managing a single listing and those managing multiple listings.
- Validation of numeric features. Variables including price, minimum nights, reviews per month, and availability were inspected for invalid entries and inconsistent formatting. These variables were retained for further processing, with outliers and extreme values addressed in the dedicated outlier treatment stage.
- Conversion of location and capacity variables. Latitude and longitude were validated as numeric coordinates, and variables such as `beds` and `bedrooms` were reviewed for consistency to ensure they correctly reflected listing capacity.
- Preparation of the cleaned dataset for further processing. The dataset was examined for structural completeness and alignment with downstream analytical

requirements. All fields were confirmed to be appropriately formatted for statistical analysis, regression modeling, classification algorithms, and clustering methods.

These cleaning steps yielded a consistent and reliable dataset that formed the basis for the subsequent stages of missing value handling, outlier treatment, feature engineering, and exploratory analysis.

• **Handling Missing Values**

A careful assessment of missing values was conducted to ensure that all analyses and models were based on complete and accurate information. Missing values were minimal in most fields, but a few variables required deliberate handling to maintain the reliability of statistical tests and machine learning procedures. The following steps summarize how missing values were addressed.

- Identification of missing entries in all dataset fields. This included review of numeric variables such as price, bedrooms, bathrooms, reviews per month, and rating, as well as categorical variables such as room type and borough. This initial assessment quantified missingness and allowed determination of which fields were essential for modeling.
- Removal of rows with missing values in critical analytical fields. A small number of entries exhibited missing information in variables necessary for modeling and inference. Rows missing values in price, borough, room type, latitude, or longitude were removed because these fields are required for price modeling, demand analysis, and spatial interpretation. The number of removed rows was small relative to the dataset size.
- Retention of missing values in non critical fields. Variables such as rating and bathrooms contained missing values that did not prevent their use in the study. These fields were left with missing entries at this stage because imputations or transformations were applied later during model specific preprocessing.
- Validation of data completeness for core analytical variables. After removal of rows missing critical information, the dataset was re examined to ensure that all variables required for regression modeling, statistical tests, classification algorithms, and clustering had sufficient completeness for reliable analysis.

This process ensured that essential fields were fully available for statistical modeling, while preserving the integrity of variables that could be reasonably imputed or transformed in later stages. The dataset retained nearly all original observations after minimal removals, resulting in a complete foundation for subsequent analysis.

• **Outlier Treatment**

Outlier treatment was an important step in preparing the dataset for reliable statistical testing and predictive modeling. Several variables exhibited extreme values that could distort averages, influence model coefficients, or reduce the stability of clustering algorithms. Because Airbnb listing characteristics often vary widely, outlier filtering was conducted in a measured and methodical way to preserve realistic observations while removing values that were incompatible with analytical assumptions. The following actions summarize the treatment of outliers.

- Review of price values to identify unrealistic nightly rates. The distribution of price revealed a long right tail with a small number of listings priced far above typical market ranges. To maintain analytical stability and prevent undue influence from extreme observations, the price variable was trimmed to retain listings with nightly rates between ten dollars and one thousand dollars. This range preserved the vast majority of listings and removed values that were not representative of the New York City market.
- Examination of minimum nights to detect implausibly large booking requirements. The minimum nights variable occasionally contained values exceeding one thousand nights. These extreme values are not realistic for short term rentals and would negatively affect statistical measures. Minimum nights was therefore constrained to a practical range of one to three hundred sixty five nights.
- Inspection of the reviews per month field to identify excessively large review rates. The reviews per month variable exhibited a heavily skewed distribution with occasional extremely high values. Listings with more than ten reviews per month were removed from the analysis because such values reflected rare or atypical activity rather than generalizable demand patterns.
- Confirmation that outlier filtering preserved dataset size and representativeness. After applying outlier thresholds for price, minimum nights, and reviews per month, the dataset retained more than ninety nine percent of original listings. This ensured that all analytical methods operated on realistic values without compromising the statistical power or representativeness of the sample.

Outlier treatment produced stable and well behaved distributions for key variables and ensured that downstream analyses, including regression, classification, and clustering, would not be unduly influenced by extreme or non representative values.

• **Feature Engineering & Variable Transformation**

Feature engineering and variable transformation were essential steps in preparing the dataset for statistical modeling, classification algorithms, and clustering procedures. Several variables required transformation to reduce skewness, improve interpretability, or create analytical categories that did not exist in the raw data. The following steps summarize the engineering and transformation procedures applied to the dataset.

- Creation of a log transformed price variable to address right skewness. The distribution of nightly price was heavily skewed, which can violate regression assumptions and reduce model stability. A log transform was applied using the function

$$\log_price = \log(1 + price)$$

to compress extreme values and support linear modeling.

- Construction of a host type variable to reflect host strategy. Hosts managing only one listing were labeled as Single and hosts managing more than one listing were labeled as Multi. This engineered feature allowed the study to examine differences in pricing behavior and guest engagement across host types.
- Conversion of bathrooms and bedrooms into numeric formats. The raw fields included text based entries such as descriptive counts or categorical labels. These were standardized and converted into numeric values to enable their use in regression models and clustering procedures.
- Creation of log base ten transformed engagement variables. Guest engagement measured through reviews per month was highly skewed. To make interpretation more meaningful and reduce the influence of extreme values, two transformed variables were created:

$$\log10_rpm = \log_{10}(1 + reviews_per_month)$$

and

$$\log10_engagement_adj = \log_{10}(1 + engagement_adj)$$

where the adjusted engagement metric accounted for availability through

$$engagement_adj = \frac{reviews_per_month}{\max\left(\frac{availability_365}{365}, 0.05\right)}$$

- Encoding of categorical variables for analytical compatibility. Categorical fields such as room type and borough category were checked for consistent categories. These variables were prepared for modeling by ensuring each category was cleanly represented and aligned with statistical and machine learning requirements.
- Identification of variables to be standardized in later phases. To prepare for clustering and classification models that rely on distance based computations, numeric variables such as log price, rating, availability, and engagement metrics were noted for later standardization as part of model specific preprocessing.

These engineered features and transformations improved the interpretability, stability, and analytical suitability of the dataset. They also provided a consistent foundation for regression modeling, demand classification, and clustering analysis.

• **Final Prepared Dataset Summary**

After completing data cleaning, missing value handling, outlier treatment, and feature engineering, the dataset was fully prepared for exploratory analysis, statistical testing, predictive modeling, and clustering. The final prepared dataset preserved nearly all observations from the original source while ensuring that all variables were in formats suitable for analytical methods. The following conditions summarize the state of the dataset after preparation.

- All key variables related to price, host characteristics, room attributes, availability, and review activity were complete and structurally consistent.
- Numeric fields such as price, minimum nights, reviews per month, rating, bedrooms, and bathrooms were validated, cleaned, and transformed as needed to support linear modeling and machine learning procedures.
- Outlier filtering ensured that unrealistic or extreme values did not distort statistical measures or model performance. Price, minimum nights, and reviews per month were trimmed to ranges representative of the New York City Airbnb market.
- Categorical fields including room type, borough category, and engineered host type were standardized for analytical compatibility. These fields were ready for both statistical comparisons and categorical encoding within modeling pipelines.
- Newly engineered variables such as log transformed price, log base ten transformed engagement metrics, and adjusted engagement indicators were incorporated into the dataset to enable more stable and interpretable modeling results.
- A synthetic unique identifier replaced the original unreliable identifier to maintain consistency and ensure that each row in the dataset represented a distinct listing.

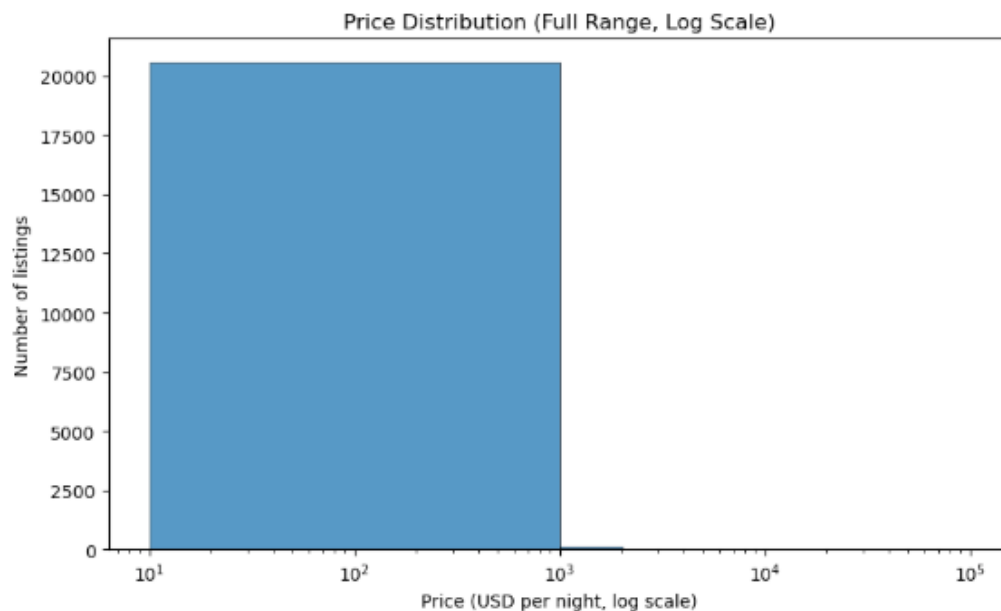
The final dataset therefore provided a solid and analytically sound foundation for comprehensive exploratory data analysis, rigorous hypothesis testing, predictive modeling, and clustering. The following chapter presents the exploratory analysis conducted to understand the structure of the data and to identify the key patterns that motivate the subsequent research questions.

⇒ Exploratory Data Analysis (EDA)

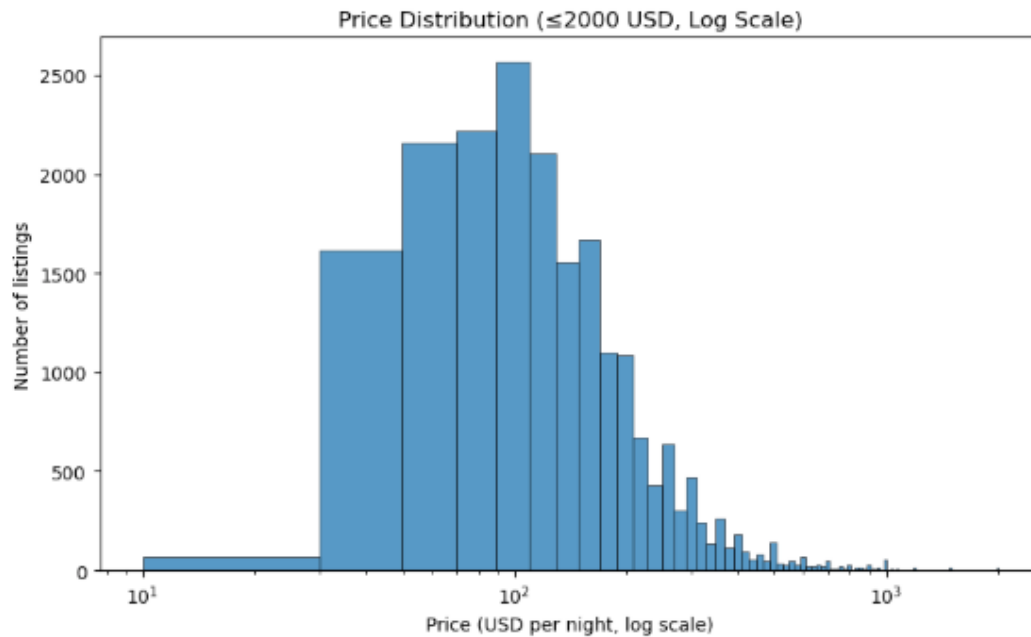
The exploratory data analysis stage provides an initial understanding of the structure of the dataset, highlights important relationships among variables, and guides the selection of appropriate statistical models. This analysis focuses on identifying patterns in price, room types, host characteristics, demand indicators, and listing availability. The following subsections summarize the principal observations from the exploratory phase.

• Price Distribution

The distribution of nightly price in the dataset revealed substantial variation across listings. The raw price values were highly right skewed, with most listings concentrated in the lower and middle ranges and a small number of extreme high priced listings. This skewness was clearly visible in the full range histogram, where a small group of listings extended the distribution far to the right. Such a distribution is common in urban short term rental markets where luxury properties or entire apartments can command very high nightly rates.



To improve interpretability and highlight the dominant structure of the data, a capped price distribution was examined by restricting the horizontal axis to values at or below two thousand dollars. Under this view, the main pricing patterns became clear. Most listings were clustered between fifty and three hundred dollars per night, with a gradual tail extending toward higher values. A boxplot of the capped distribution further illustrated the presence of outliers and the relatively narrow interquartile range for typical listings.

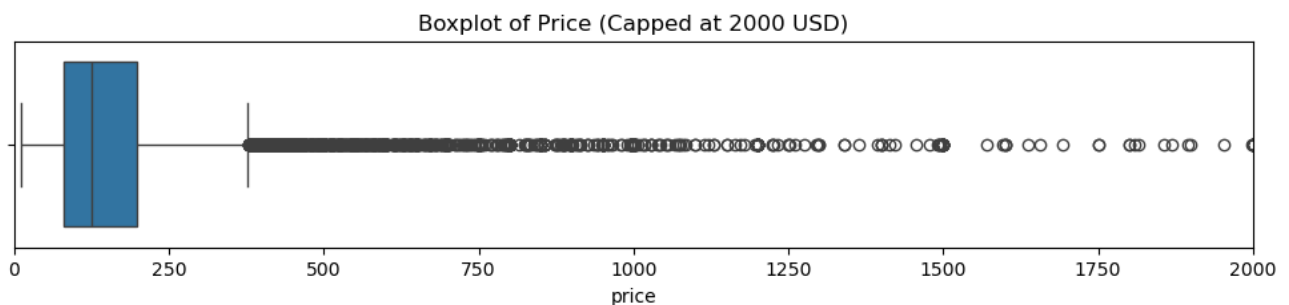


The skewed distribution of price motivated the use of a log transformed price variable in regression modeling. The transformation

$$\log_price = \log(1 + price)$$

helped reduce skewness and produced a more symmetric distribution suitable for linear modeling. This transformation preserved the relative ranking of listings while stabilizing variance across the price spectrum.

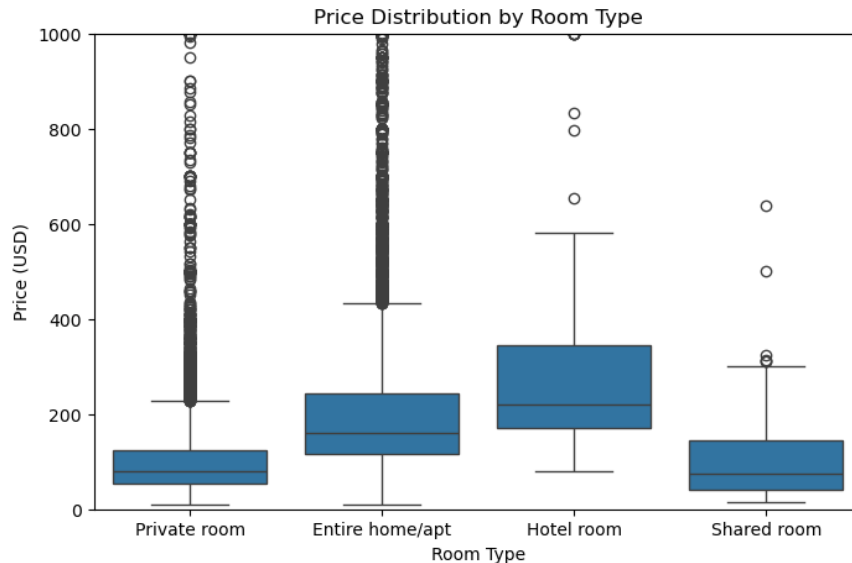
The exploratory analysis of nightly price therefore confirmed that New York City Airbnb listings exhibit wide pricing diversity and that careful treatment of the price variable is required for effective statistical modeling. These findings guided the design of the analytical methods used in the subsequent chapters.



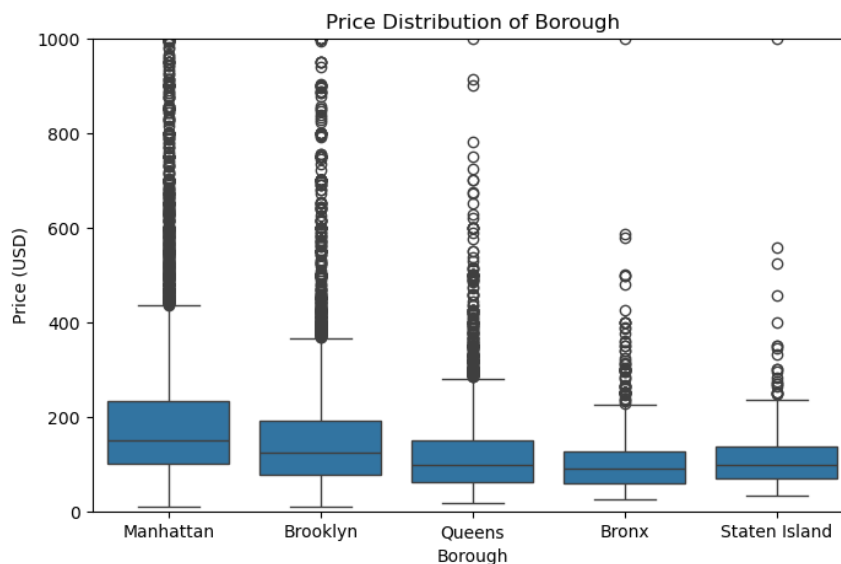
• Room Type and Borough Patterns

The relationship between price and structural listing characteristics was explored by examining how room type and borough categories influence nightly rates. These variables represent key dimensions of the New York City short term rental market because room type limits the size and privacy of the accommodation, while borough location reflects broader differences in demand, density, and neighborhood desirability.

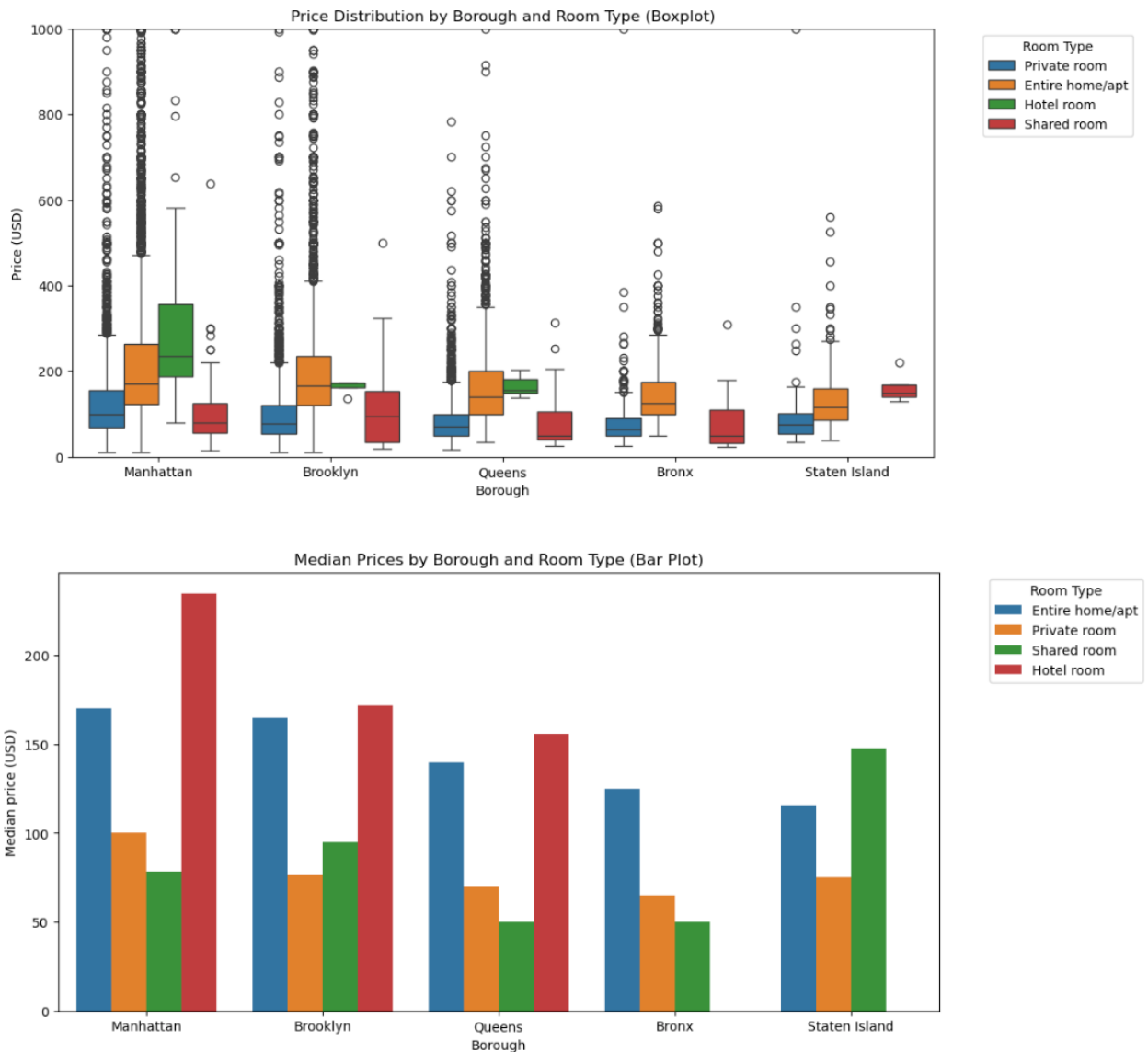
The distribution of price across room types showed clear and intuitive patterns. Entire homes and apartments had substantially higher median prices compared to private rooms and shared rooms. Hotel rooms within the Airbnb platform also exhibited elevated prices relative to shared or private accommodations. Boxplots revealed that entire homes displayed the widest range of prices, including a substantial number of high priced listings. Private rooms formed a narrower band of values, and shared rooms consistently represented the lowest pricing tier.



Borough level analysis showed similarly well defined price differences. Manhattan had the highest median nightly price, followed by Brooklyn, while listings in Queens, the Bronx, and Staten Island were more affordable. The price structure across boroughs was consistent with the well known spatial distribution of real estate values and tourism patterns in New York City. Boxplots further illustrated the wide dispersion of prices in Manhattan and Brooklyn compared to the relatively tight distributions observed in the other boroughs.



To examine these variables jointly, price distributions were analyzed across combinations of room type and borough. The resulting comparisons revealed that both attributes contribute meaningfully to price formation. For example, entire homes in Manhattan and Brooklyn occupy the highest price band, whereas private rooms in Queens or the Bronx are among the most affordable options. A pivot table summarizing median price by room type and borough highlighted these combined effects clearly, and a heatmap offered a concise visual representation of the patterns.

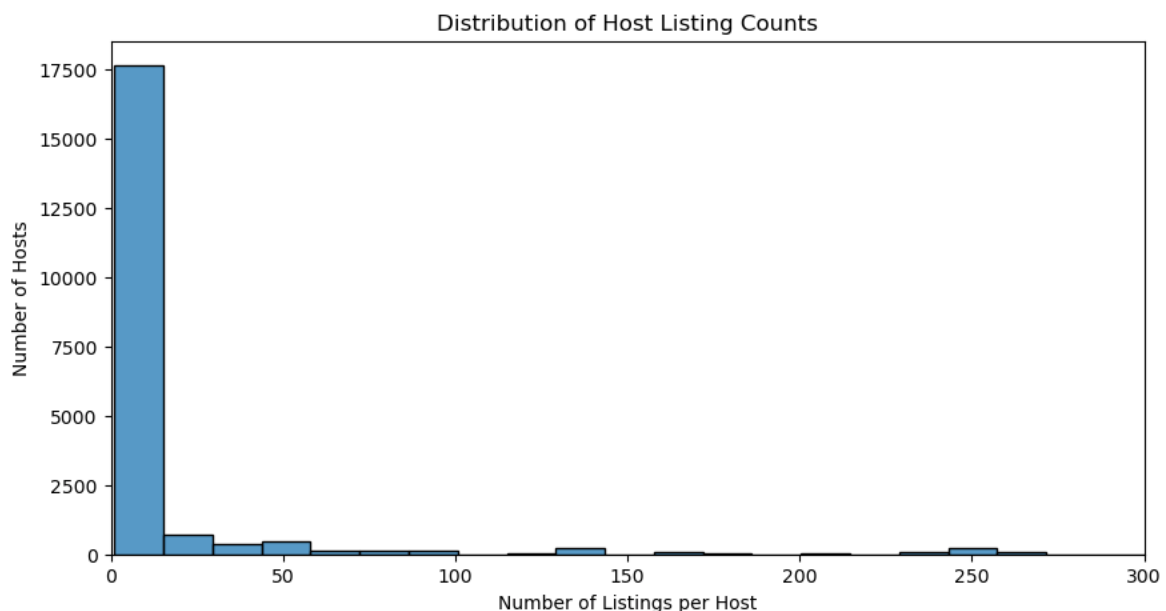


Together, the exploratory analysis of room type and borough categories demonstrated that location and accommodation type are central determinants of price variation in the New York City Airbnb market. These insights directly motivated the statistical tests and regression models presented in later chapters.

- **Host Behavior Patterns**

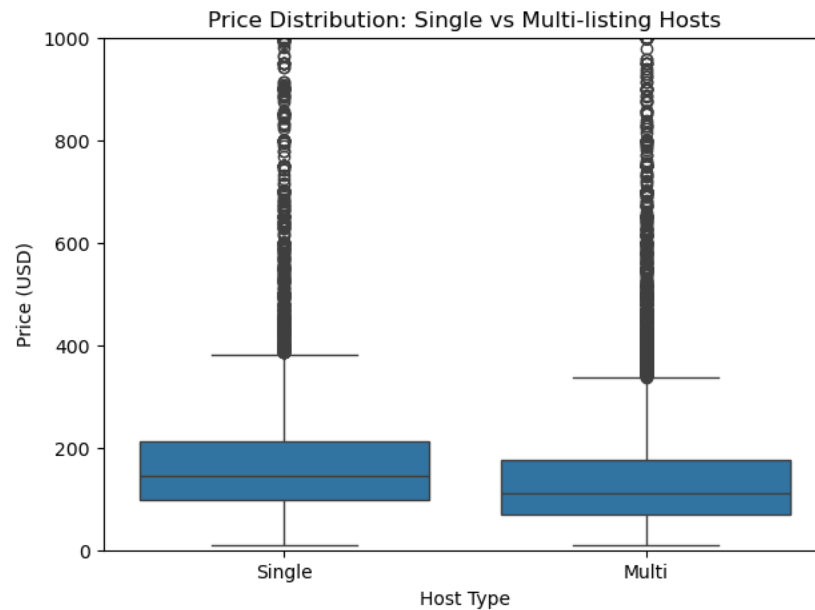
Host behavior plays an important role in shaping overall market dynamics, particularly in the context of pricing strategy, listing activity, and guest engagement. The exploratory analysis of host related variables focused on the number of listings managed by each host and the behavior of hosts who operate single listings compared to those who operate multiple listings. These insights helped establish patterns that later guided the statistical tests and regression models related to host strategy.

The distribution of listings per host revealed a strongly right skewed pattern. Most hosts managed a single listing, reflecting the participation of individual property owners in the platform. A smaller number of hosts managed several listings, and a very limited group oversaw hundreds of properties. This long tailed distribution indicated the presence of commercial style operators whose scale differs significantly from individual hosts.

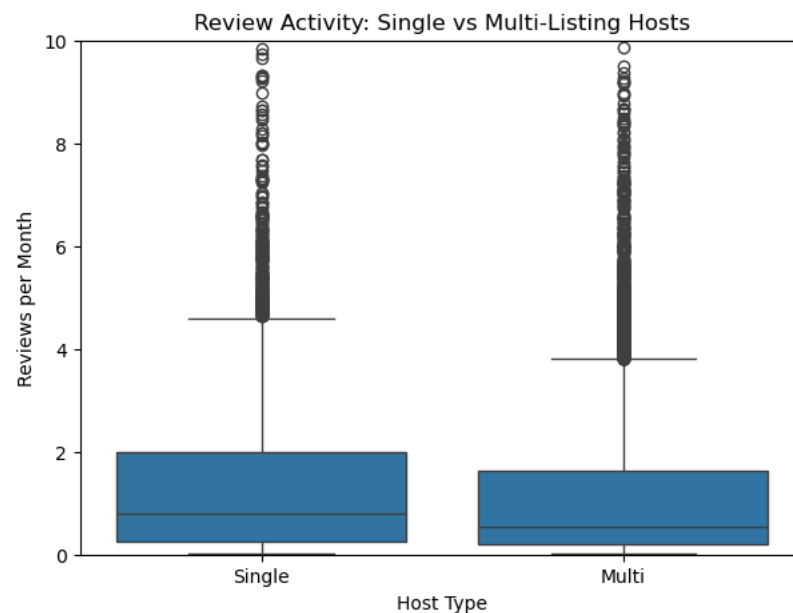


To distinguish between these behaviors, an engineered variable was created to classify hosts as either Single or Multi based on whether they managed one listing or more than one. This categorization allowed focused comparison of pricing and engagement patterns across host types.

Exploratory analysis showed that Single hosts tended to charge higher nightly prices than Multi hosts. Boxplots comparing these groups revealed that median prices for Single hosts were notably higher than for Multi hosts. This suggested that individual hosts may price their listings at a premium either because they offer more personalized accommodations or because they operate in higher demand areas.



Guest engagement also varied across host types. Single hosts exhibited slightly higher values for reviews per month compared to Multi hosts, indicating that they may receive more consistent guest feedback or operate listings with more frequent bookings. However, this raw difference was interpreted with caution, because engagement is influenced by listing availability and other listing characteristics, which are explored in later sections.

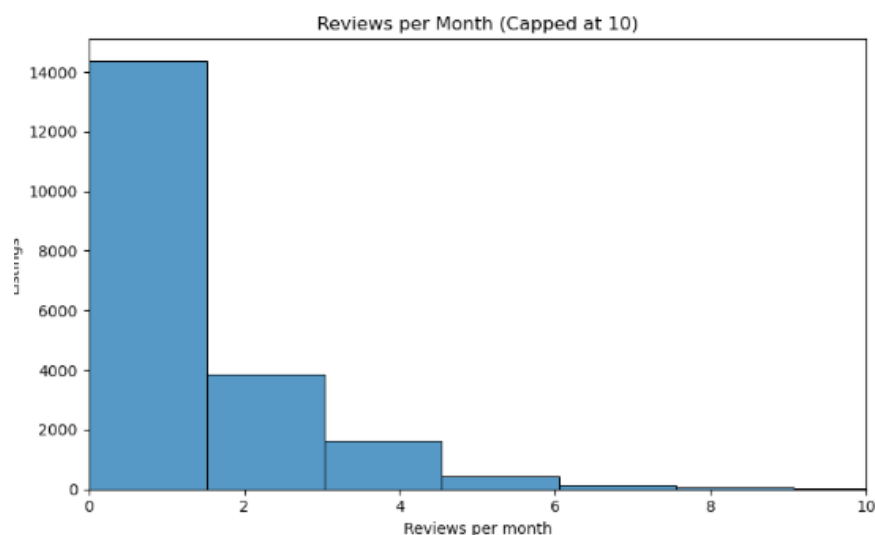
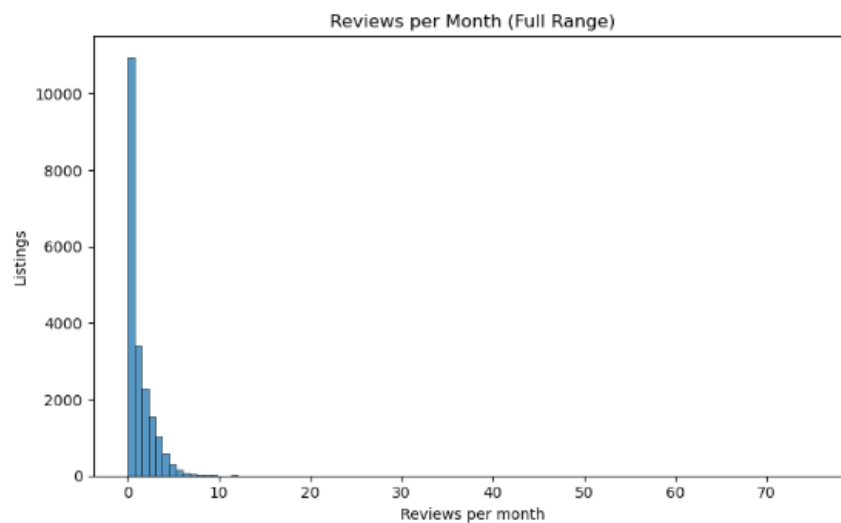


Overall, the exploratory analysis demonstrated that host strategies differ meaningfully within the platform. The distinction between Single and Multi hosts provided a foundation for the formal hypothesis testing and regression modeling conducted in subsequent chapters to determine whether these observed differences were statistically significant.

- **Review Activity and Demand Patterns**

Review activity serves as a useful proxy for understanding demand patterns within the Airbnb marketplace. The variable representing reviews per month captures how frequently guests interact with and respond to individual listings, and therefore provides insight into booking frequency and listing relevance. The exploratory analysis of review patterns focused on the distribution of reviews per month, the identification of high demand listings, and the relationships between review activity and other listing characteristics.

The distribution of reviews per month exhibited a strongly right skewed shape. Most listings received fewer than one review per month, while a smaller fraction accumulated reviews more frequently. The presence of skewness reflected the influence of a limited number of high turnover listings that receive guest feedback at an elevated rate. Visualizing the distribution made it clear that review activity varies widely across listings and cannot be modeled effectively without transformation. This motivated the use of a log base ten transformation of reviews per month during modeling phases to stabilize variance and improve interpretability.



To understand differences in review activity across host strategies, the distribution of reviews per month was compared between Single and Multi hosts. Single hosts exhibited slightly higher engagement levels than Multi hosts in the raw data, suggesting that their properties may attract more guest interactions. However, these findings were interpreted cautiously because review activity is influenced not only by host strategy but also by listing price, availability, and room type.

Availability was also considered during the exploratory analysis because it directly affects a listing's opportunity to receive reviews. Listings that were available for most of the year did not always exhibit high review activity, while listings with limited availability sometimes received large numbers of reviews. This pattern suggested that availability alone does not determine demand, and that review activity is shaped jointly by pricing, listing features, and host behavior.

The exploratory analysis of review activity highlighted the complexity of demand dynamics in the New York City Airbnb market. The wide dispersion of reviews per month indicated that any classification of listings into high demand or low demand categories would require machine learning methods rather than simple thresholding. These observations motivated the use of classification models to identify high demand listings and to examine how listing characteristics contribute to review based demand segmentation.

- **Summary of Key EDA Insights**

The exploratory analysis revealed several important patterns that define the structure of the New York City Airbnb market. The distribution of price showed substantial right skewness, with most listings concentrated in the lower to middle price ranges and a smaller number of premium listings creating a long upper tail. When price was analyzed by room type and borough, clear differences emerged. Entire homes and hotel style listings tended to exhibit higher prices than private or shared rooms, and listings in Manhattan and Brooklyn were priced significantly higher than those in Queens, the Bronx, or Staten Island. These findings established that price formation in the market is influenced heavily by listing type and geographic location.

Host behavior contributed additional structure to the market. Most hosts managed only a single listing, while a smaller group managed many. Single hosts tended to price their listings at higher levels and showed slightly stronger review activity. Multi hosts, on the other hand, tended to operate with broader availability and at somewhat lower prices. These patterns suggested distinct host strategies that may influence guest engagement and listing performance.

Availability patterns indicated that many listings were available for a large portion of the year, while others offered more limited access. Availability alone did not determine review activity. Some listings that were available for long periods received relatively few reviews, while others with limited availability attracted more consistent

engagement. This indicated that availability interacts with price, listing characteristics, and rating quality to influence demand.

Ratings patterns revealed that most listings received high scores, creating a compressed distribution near the top of the scale. Although the variance in ratings was limited, differences in rating still played a measurable role in demand. Listings with higher ratings tended to receive more reviews per month and appeared to have stronger guest engagement. This provided motivation to include ratings in modeling exercises that aimed to identify drivers of engagement and demand.

Overall, the exploratory analysis demonstrated that price, room type, borough location, host strategy, availability, and ratings collectively shape the behavior of listings in the New York City Airbnb market. These insights informed the development of the research questions and guided the selection of appropriate statistical tests and modeling techniques used in subsequent chapters.

⇒ Research Questions Overview

The exploratory analysis revealed meaningful structural patterns in price, host behavior, guest engagement, availability, and ratings. These observations motivated a set of research questions designed to guide the statistical analysis, predictive modeling, and segmentation work conducted in this study. The research questions synthesize the central themes identified during data exploration and provide a coherent structure for investigating the economic and behavioral dynamics of Airbnb listings in New York City.

The first research question examines how prices differ across boroughs and room types. These two variables showed strong influence on nightly price in the exploratory analysis, and understanding the magnitude and statistical significance of these differences is essential for characterizing market structure.

The second research question focuses on identifying the characteristics that most strongly predict price. Because price is a key determinant of listing performance, a regression based analysis was used to quantify the relative influence of room type, location, property size, rating, and host behavior.

The third and fourth research questions investigate host strategy. The third question asks whether multi listing hosts set different prices from single listing hosts, while the fourth question examines whether host strategy influences guest engagement as measured by review activity. These questions aim to clarify how host behavior affects both price formation and demand.

The fifth research question evaluates whether listings can be classified into high demand and low demand segments based on review patterns. The presence of skewed review distributions suggested that demand varies significantly across listings, and a classification based approach was applied to identify the features that differentiate high demand listings from others.

The sixth research question explores whether listings can be grouped into clusters based on price, host behavior, ratings, and reviews. This unsupervised learning task identifies natural market segments and provides insight into how combination of factors jointly define distinct listing profiles within the city.

Together, these research questions form a structured analytical roadmap that links the exploratory findings to the statistical methods and results presented in the next chapters.

⇒ Methodology

The analytical approach used in this study combines inferential statistics, regression modeling, classification techniques, and clustering. Each component was selected to address specific research questions and to match the structural characteristics of the dataset. The methods described in this chapter form the foundation for the empirical results presented later.

• Statistical Tests

Statistical tests were applied to assess whether observed differences in price and engagement across groups were statistically significant. These tests provided formal support for conclusions drawn from the exploratory analysis and guided the interpretation of the regression and classification results. The tests used in this study include classical parametric approaches, non parametric alternatives, and robust procedures for ordinal and skewed data.

The first set of tests examined price differences across boroughs and room types. Because price distributions were right skewed even after trimming extreme values, log transformed price was used when appropriate. Two types of tests were conducted. The first was analysis of variance, expressed as

$$F = \frac{\text{Between Group Variance}}{\text{Within Group Variance}}$$

which assesses whether mean log prices differ across categories. After testing assumptions regarding normality and variance homogeneity, Welch analysis of variance was used for cases where group variances differed. The second test was the Kruskal Wallis statistic, defined by

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k n_i (R_i - \bar{R})^2$$

which compares ranked outcomes and does not require normality. This test was selected because the price variable remained skewed even after transformation.

To evaluate differences in price between Single and Multi hosts, two sample comparisons were performed. The Mann Whitney U statistic, defined by

$$U = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1$$

was used to compare distributions of price and review based engagement metrics across host types. This non parametric test was appropriate because engagement variables such

as reviews per month were highly skewed and contained outliers even after transformation.

These statistical tests provided formal evidence regarding group differences and established whether patterns observed in exploratory analysis represented meaningful and statistically significant effects. The results of these tests are presented in later chapters.

- **Regression Modeling**

Regression modeling was used to quantify the influence of listing characteristics on price and guest engagement. These models addressed questions related to price prediction, host strategy effects, and the relationship between listing features and review activity. Regression methods provided a structured quantitative framework for isolating the contribution of individual variables while controlling for multiple covariates simultaneously.

Two primary forms of regression models were applied. The first was ordinary least squares regression used to model log transformed price. Nightly price was right skewed in its raw form, and a log transformation was applied to stabilize variance and support linearity assumptions. The transformed dependent variable is defined as

$$\log_price = \log(1 + price)$$

This allowed the model to capture relative rather than absolute price differences and reduced the influence of extreme observations. The model included room type, borough category, host type, beds, bedrooms, bathrooms, rating, and availability as predictors. These predictors were selected based on insights from the exploratory analysis and their theoretical relevance to pricing behavior.

The second set of regression models focused on guest engagement through review activity. Because review counts per month were highly skewed, a log base ten transform was used to improve interpretability and reduce the impact of extreme values. The dependent variable for modeling raw engagement was

$$\log_{10}_rpm = \log_{10}(1 + reviews_per_month)$$

and the dependent variable for modeling engagement adjusted for availability was

$$\log_{10}_engagement_adj = \log_{10}(1 + engagement_adj)$$

These models included the same set of predictors as the price model, allowing direct comparison between the determinants of price and the determinants of engagement.

Host strategy was incorporated into the regression framework through the engineered host type variable. This allowed the analysis to measure whether hosts managing single

listings behaved differently from those managing multiple listings. The regression approach provided estimates of the marginal effect of host type on price and on engagement after controlling for location, room type, property size, rating, and availability.

All regression models were estimated using robust standard errors to mitigate the effects of heteroskedasticity. Model diagnostics, including residual plots, normal probability plots, and assessments of variance stability, were used to verify the suitability of the linear modeling assumptions. These diagnostics supported the use of linear regression for log transformed price and for the log transformed review based metrics.

Regression modeling therefore provided a principled statistical approach for answering research questions related to price formation and guest engagement. The results derived from these models are presented in later chapters.

• **Classification Modeling**

Classification modeling was used to address the research question concerning whether listings can be categorized into high demand and low demand segments based on review activity and listing characteristics. The motivation for this approach stemmed from the skewed distribution of reviews per month and the observation that certain listings consistently received more guest interactions than others. Classification methods allowed the analysis to formalize this distinction and identify the factors that best predict demand level.

The target variable for classification was created by dividing listings into two groups based on the median value of reviews per month. Listings with review activity at or above the median were classified as high demand, and listings below the median were classified as low demand. This binary segmentation approach ensured a balanced distribution across the two categories and provided a clear outcome variable for supervised learning models.

Several classification algorithms were applied in order to capture both linear and nonlinear relationships among predictors. Logistic regression served as the baseline model. In this model, the probability that a listing is high demand is modeled as

$$P(Y = 1 | X) = \frac{1}{1 + e^{-(\beta_0 + X\beta)}}$$

where Y is the binary demand outcome and X represents the listing features. Logistic regression provided a straightforward interpretation of how each variable influences the likelihood of high demand and established a benchmark for evaluating more flexible models.

Random forest classification was used to capture nonlinear interactions between variables. Random forests aggregate predictions from a large number of decision trees, each constructed from bootstrap samples, and use majority voting for classification. This approach is well suited for heterogeneous datasets that include both numeric and categorical variables. Random forest classification allowed the analysis to quantify variable importance and evaluate the extent to which complex interactions drive demand.

The third classification model used was the histogram based gradient boosting classifier. This method builds decision trees sequentially, with each tree focusing on prediction errors made by the previous ones. Histogram based gradient boosting is highly efficient for large datasets and is capable of modeling subtle nonlinearities and interactions. It served as the most advanced model in the classification pipeline and provided the strongest performance.

All classification models were evaluated using accuracy, precision, recall, and the area under the receiver operating characteristic curve. Confusion matrices were used to assess the distribution of correct and incorrect predictions across the high demand and low demand categories. The models were compared to determine the extent to which linear and nonlinear relationships influence demand classification.

The classification modeling framework provided systematic insight into how price, availability, rating, host strategy, and property characteristics contribute to demand segmentation. The results of these models are presented in later chapters.

- **Clustering Method**

Clustering was used to identify natural market segments within the Airbnb listings based on multiple listing characteristics that jointly influence price, demand, and host behavior. This unsupervised learning approach allowed the study to uncover patterns that are not directly observable through individual variables or supervised models. Clustering was specifically applied to answer the research question concerning whether listings can be grouped into statistically similar market segments based on their aggregated characteristics.

The clustering analysis used a set of continuous variables that represent key dimensions of listing performance. These included log transformed price, rating score, reviews per month, availability, and a binary indicator for host type. This combination of variables incorporates information about supply side characteristics, demand side engagement, host strategy, and listing quality. Before applying clustering, all numeric variables were standardized so that each variable contributed equally to the distance based algorithm. Scaling was performed because K means clustering relies on Euclidean distance, which is sensitive to differences in variable magnitude.

The K means algorithm partitions data into a fixed number of clusters by minimizing the sum of squared distances between each observation and the centroid of its assigned cluster. Formally, the algorithm seeks to minimize

$$\sum_{i=1}^k \sum_{x \in C_i} |x - \mu_i|^2$$

where k is the number of clusters, C_i represents the set of observations assigned to cluster i , and μ_i is the centroid of that cluster. The iterative procedure alternates between assigning observations to the nearest centroid and updating centroid positions until convergence is achieved.

The optimal number of clusters was determined using the Elbow Method, which evaluates the within cluster sum of squares across a range of values for k . The plot of within cluster variability against the number of clusters typically decreases as k increases, but exhibits a point where the rate of improvement begins to slow. This inflection point, often called the elbow, provides a guide for selecting k . In this study, the elbow occurred at four clusters, and this value was selected for the final segmentation analysis.

To support interpretation of cluster structure, principal component analysis was used to project the standardized variables into two dimensions for visualization. Principal component analysis does not influence the clustering solution but provides a means to visually examine the separation between clusters and to interpret how listings group together in reduced dimensional space.

The clustering method therefore enabled the identification of distinct market segments based on the combined influence of price, host behavior, quality ratings, review patterns, and availability. These segments help characterize the structure of the New York City Airbnb market and are discussed in the results chapter.

⇒ Results

The results presented in this chapter address each research question using the statistical tests, regression models, classification methods, and clustering techniques described in the methodology chapter. The findings provide a comprehensive understanding of pricing, host behavior, guest engagement, demand segmentation, and market structure in the New York City Airbnb market.

• **RQ1: How do nightly prices differ across boroughs and room types?**

This research question examines whether nightly prices vary significantly across boroughs and room types. The exploratory analysis showed clear price differences across both categories, and the statistical tests confirmed that these differences are substantial and statistically significant. The results presented below summarize these findings using descriptive evidence, visual patterns, and inferential test statistics.

Nightly price exhibited strong variation across room types. The boxplot of price by room type showed consistent separation between categories, with entire homes and hotel style listings positioned well above private and shared rooms. Median values highlighted the magnitude of these differences.

Room type effects:

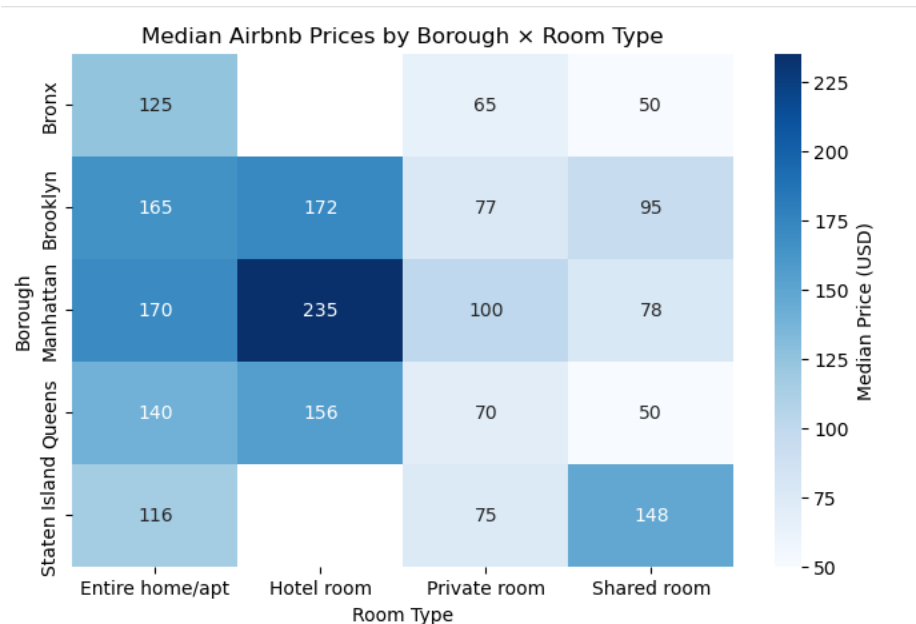
- Median price for **Entire homes**: approximately 160 USD
- Median price for **Hotel rooms**: approximately 221 USD
- Median price for **Private rooms**: approximately 80 USD
- Median price for **Shared rooms**: approximately 75 USD

Price variation across boroughs followed an equally clear pattern. The boxplot for boroughs showed that listings located in Manhattan and Brooklyn consistently commanded higher nightly prices than those in Queens, the Bronx, or Staten Island.

Borough effects :

- **Manhattan**: median price 150 USD
- **Brooklyn**: median price 124 USD
- **Queens**: median price 98 USD
- **Staten Island**: median price 99 USD
- **Bronx**: median price 89 USD

Price differences became even more pronounced when room type and borough were examined jointly. The pivot table summarizing median prices across both dimensions showed a clear hierarchy of pricing levels. A heatmap illustrated the same pattern visually, with darker cells corresponding to the highest priced listing categories.



Combined room type and borough effects:

- Highest priced segment: **Manhattan hotel rooms** with a median price of **approximately 235 USD**
- High priced segment: **Manhattan entire homes** with median prices above **170 USD**
- Lowest priced segments: **Private and shared rooms** in **Queens** and **the Bronx**, with median prices between **50 and 100 USD**

To statistically confirm these observed differences, log transformed nightly price was analyzed using analysis of variance, Welch analysis of variance, and the Kruskal Wallis test.

Statistical tests for borough differences:

- **ANOVA:** $F(4, 20481) = 416.5, p < 0.001$
- **Welch ANOVA:** statistic **443.7, $p < 0.001$**
- **Kruskal Wallis:** $H=1553.6, p < 0.001$

Statistical tests for room type differences:

- **ANOVA:** $F(3, 20482) = 2525.0, p < 0.001$
- **Welch ANOVA:** statistic **approximately 2461.0, $p < 0.001$**
- **Kruskal Wallis:** $H=5979.6, p < 0.001$

All three testing approaches demonstrated highly significant group differences, confirming that the variation in nightly price across boroughs and room types is not random and instead reflects genuine structural differences in the market.

In summary, room type and borough exert strong and systematic influence on nightly price in the New York City Airbnb market. Entire homes and hotel style listings are priced substantially higher than private and shared rooms, and Manhattan and Brooklyn dominate the upper end of the price distribution. These results establish the importance of location and accommodation type, which are incorporated as primary predictors in the regression models discussed in the next research question.

- **RQ2: What property and location characteristics are the strongest predictors of price?**

This research question examines which listing characteristics most strongly influence nightly price. Linear regression using log transformed price and a histogram based gradient boosting model were applied to identify key predictors. The analysis incorporated room type, borough, host strategy, property size, rating, and availability as covariates. Diagnostic plots were used to evaluate model assumptions and assess model fit.

The distribution of nightly price was initially right skewed. To stabilize variance and support linear regression assumptions, price was transformed using the function

$$\log_price = \log(1 + price)$$

The transformed variable exhibited a more symmetric distribution, which supported the linear modeling framework.

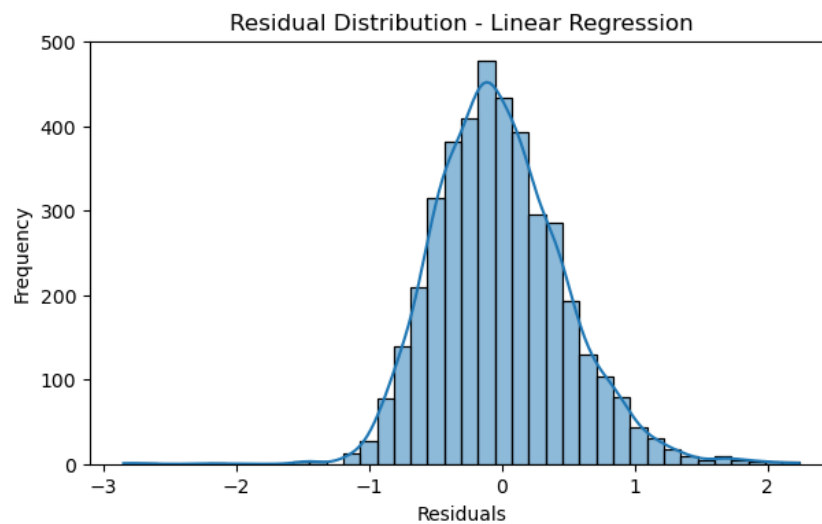
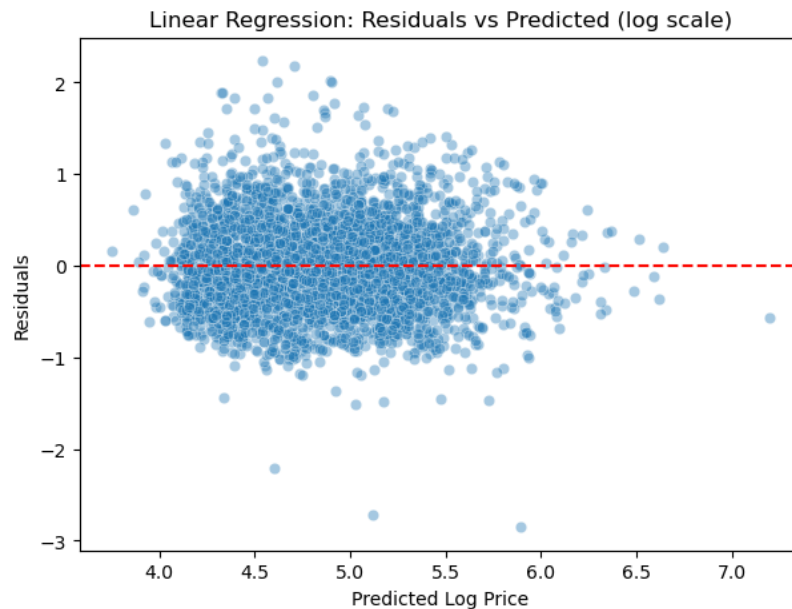
The regression model produced several coefficients that were statistically significant and economically meaningful. These coefficients identified the listing characteristics that most strongly predict price.

Key regression findings (based on log transformed price):

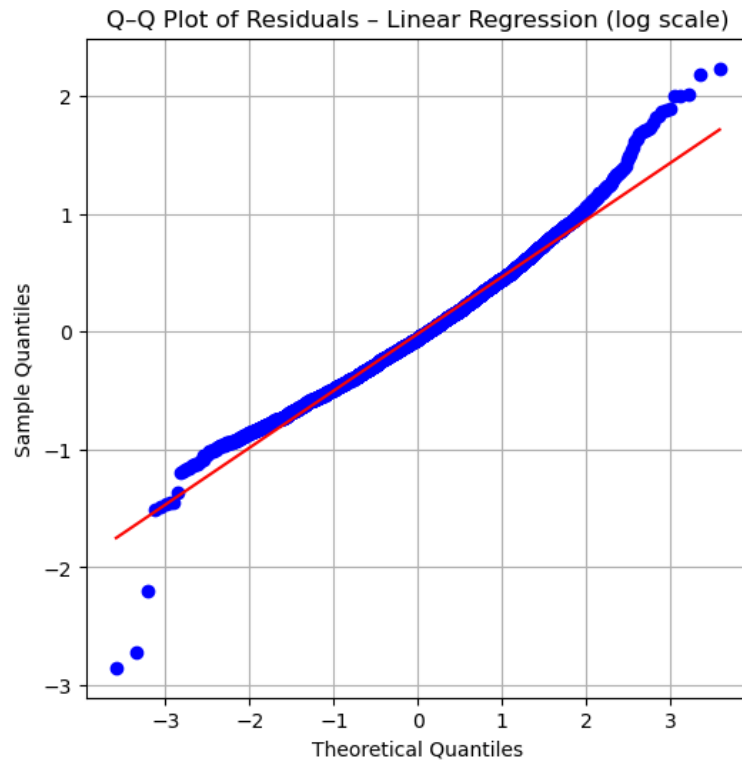
- **Room type effects**
 - Private rooms were priced **approximately 37 percent lower** than entire homes.
 - Shared rooms were priced **approximately 49 percent lower** than entire homes.
 - Hotel rooms were priced **approximately 52 percent higher** than entire homes.
- **Borough effects**
 - Manhattan listings were priced **approximately 59 percent higher** than the baseline borough.
 - Brooklyn listings were priced **approximately 22 percent higher**.
 - Queens, the Bronx, and Staten Island showed smaller positive effects.
- **Property size effects**
 - Each additional bedroom increased price by **approximately 14 percent**.
 - Each additional bed increased price by **approximately 10 percent**.
 - Each additional bathroom increased price by **approximately 9 percent**.
- **Quality and host strategy effects**
 - A one unit increase in rating increased price by **approximately 22 percent**.
 - Single hosts charged **approximately 20 percent higher prices** than Multi hosts.

Regression Diagnostics and Interpretation

Regression diagnostics were examined to confirm model stability and appropriateness. The residuals versus predicted plot showed that residuals were centered around zero, with no strong curvature or evidence of model bias. This provided support for the linearity assumption.



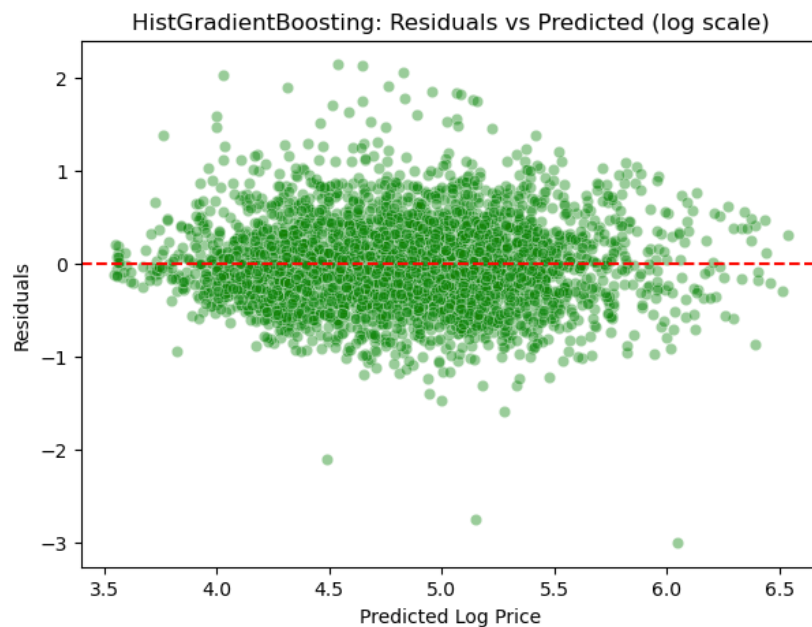
The residual distribution plot showed a bell shaped curve in the residuals, indicating approximate normality. To evaluate this more formally, a normal probability plot was create. Points were closely aligned with the theoretical line, with mild deviations at the tails. This confirmed that the log transformation improved normality and that the residual distribution was sufficiently stable for inference.

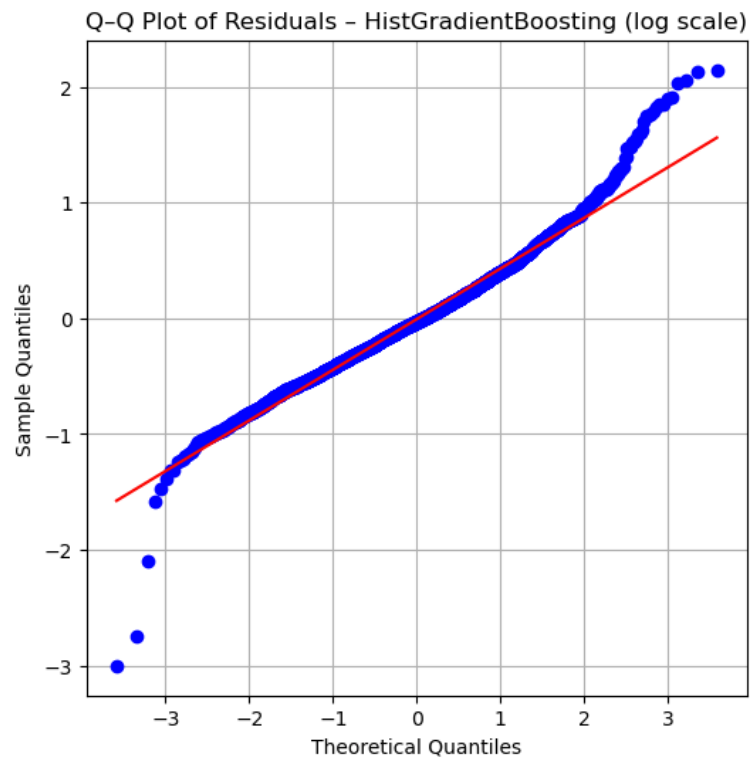
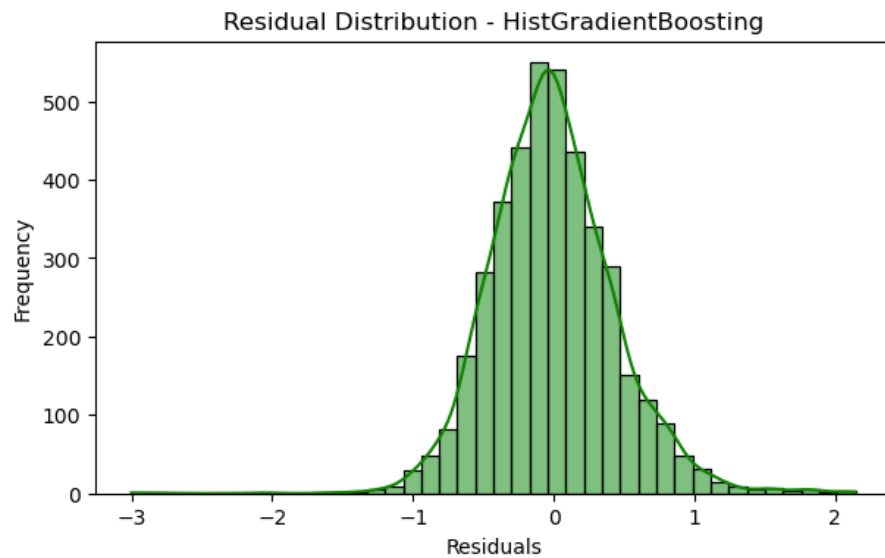


These diagnostics collectively indicated that the regression model satisfied key linear modeling assumptions and that coefficient estimates were reliable.

Nonlinear Model Confirmation

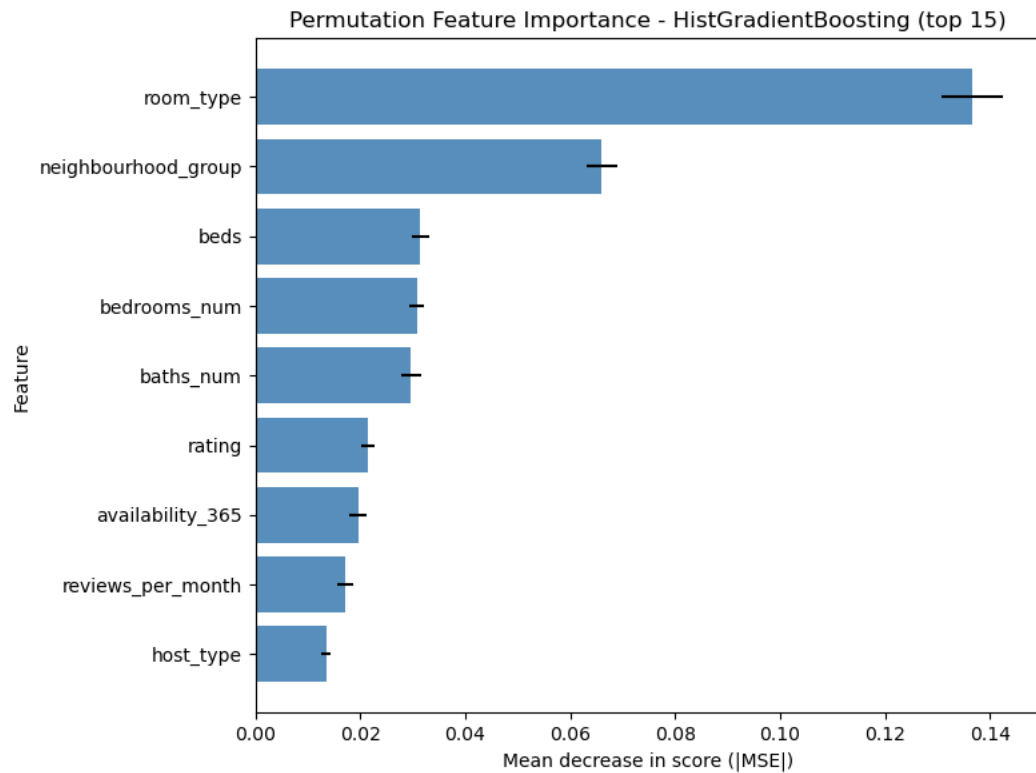
To complement the linear regression, a histogram based gradient boosting model was used to capture nonlinear relationships and potential interactions among variables. The model achieved a higher predictive accuracy and provided feature importance measures shown in Figure.





Top predictors from the nonlinear model:

- Room type
- Borough
- Beds and bedrooms
- Rating
- Host type



These variables appeared at the top of the importance ranking, confirming the findings from the regression model. The clustering of importance scores around these features indicated that price is primarily determined by supply side attributes including property type, location, capacity, and perceived quality.

Model Performance Metrics

To evaluate how effectively the regression models predicted log transformed nightly price, two key performance metrics were computed. The first was the coefficient of determination R^2 , which measures the proportion of variance explained by the model. The second was the root mean squared error (RMSE), which measures the typical magnitude of prediction error on the log scale. These metrics were calculated for both the linear regression model and the histogram based gradient boosting model.

Model	(R^2)	RMSE (log scale)	Interpretation
Linear Regression	0.447	0.488	Explains approximately 45 percent of variation in log price. Indicates moderate predictive capability with linear relationships.
HistGradientBoosting	0.546	0.442	Explains approximately 55 percent of variation in log price. Provides stronger predictive performance and captures nonlinear effects missed by the linear model.

These results show that the histogram based gradient boosting model delivers improved accuracy relative to the linear model. The improvement of approximately ten percentage

points in R^2 indicates that nonlinear interactions among room type, borough, listing size, and rating contribute meaningfully to price formation. The reduction in RMSE further confirms that the boosted model predicts nightly price more precisely.

The performance metrics therefore support the use of tree based ensemble methods alongside linear regression. While the linear model provides interpretability through coefficients, the nonlinear model captures complex structural relationships within the dataset. Together, these models offer a comprehensive understanding of how listing characteristics influence price.

Overall Interpretation for RQ2

The results demonstrate that nightly price is strongly influenced by room type and borough location, with property size, rating, and host strategy playing secondary but meaningful roles. Regression diagnostics confirmed that the linear model was appropriate for the transformed price variable, and the nonlinear gradient boosting model corroborated the importance of these predictors. Together, these findings show that price formation in the New York City Airbnb market is driven primarily by structural listing characteristics and geographic positioning.

- **RQ3: Do multi listing hosts set different prices compared to single listing hosts?**

This research question examines whether hosts who manage multiple listings follow different pricing strategies compared to hosts who manage only one listing. The exploratory analysis revealed initial differences between the two host types, and both non parametric testing and regression modeling were applied to determine whether these differences were statistically significant.

A boxplot comparing nightly prices for Single and Multi hosts showed that Single hosts typically charge higher nightly rates. The median nightly price for Single hosts was approximately **145 USD**, while Multi hosts had a median near **110 USD**. These differences suggested that Single hosts place their listings in a higher pricing tier than Multi hosts.

To formally test this observation, a Mann Whitney U test was conducted to compare the price distributions of the two independent groups. The test yielded

$$U = 62746750, \quad p < 0.001,$$

indicating that the difference in nightly prices between Single and Multi hosts is statistically significant. Because the Mann Whitney U test does not rely on normality assumptions, it is appropriate for the skewed price distribution observed in the dataset.

A summary of these key results is presented in Table, which reports the median prices for each host group, the U statistic, and the associated p value. The table provides a concise overview of the magnitude and significance of the price differences.

Variable	Coefficient	Significance	Interpretation
Host Type (Single)	+0.159	p < 0.001	Single hosts charge ≈17% higher prices than multi hosts (holding other factors constant).
Room Type (Hotel room)	+0.459	p < 0.001	Hotel rooms are ≈58% more expensive than entire homes.
Room Type (Private room)	-0.432	p < 0.001	Private rooms are ≈35% cheaper .
Borough (Manhattan)	+0.428	p < 0.001	Manhattan listings ≈53% more expensive than baseline.
Beds / Bedrooms / Baths	+0.10–0.14	p < 0.001	Larger listings command higher prices.
Rating	+0.216	p < 0.001	Better-rated listings are priced higher.

In addition to distributional analysis, regression modeling was used to evaluate host strategy while controlling for location, room type, property size, rating, and availability. The log transformed price model included host type as a categorical predictor. The coefficient for Single hosts was **0.159**, which corresponds to a percentage change of

$$(e^{0.159} - 1) \times 100 \approx 17.2\%$$

This indicates that Single hosts charge approximately **17 percent higher** nightly prices than Multi hosts after controlling for other listing characteristics. The effect was statistically significant with a p value less than 0.001.

Key numerical results:

- Median nightly price for **Single hosts: 145 USD**
- Median nightly price for **Multi hosts: 110 USD**
- Mann Whitney U statistic: **62,746,750**
- p value: **< 0.001**
- Regression coefficient for Single hosts: **0.159**
- Adjusted percentage effect: **17.2 percent higher prices**

These findings demonstrate that host strategy is linked to pricing behavior in the New York City Airbnb market. Single hosts consistently charge higher prices, likely reflecting the unique or personalized positioning of their listings, whereas Multi hosts often operate at larger scale and may prioritize occupancy over premium pricing.

In summary, both non parametric testing and regression modeling confirm that Single hosts set significantly higher nightly prices than Multi hosts. This result provides evidence that host strategy influences market outcomes in a measurable way.

• RQ4: How does Host Strategy influence guest engagement, as measured by reviews?

This research question evaluates whether hosts who manage a single listing experience different levels of guest engagement compared to hosts who manage multiple listings. Engagement is represented through review activity, which reflects how frequently guests interact with listings after completion of a stay. Because review activity is skewed and influenced by availability, both raw review frequency and availability adjusted engagement metrics were analyzed.

The exploratory analysis showed that Single hosts tended to receive slightly more reviews per month than Multi hosts. This pattern was visible in a boxplot comparing the two groups, where Single hosts exhibited a higher median review rate. The median reviews per month for Single hosts was **0.81**, compared to **0.54** for Multi hosts. This descriptive difference suggested stronger guest engagement for Single hosts.

Metric	U Statistic	p-value	Decision	Interpretation
Reviews per Month	56,303,886	< 0.001	Reject H ₀	Single hosts receive more reviews per month.
Engagement (Adj)	60,898,669	< 0.001	Reject H ₀	Single hosts have higher engagement after adjusting for availability.

A Mann Whitney U test was conducted to determine whether this difference was statistically significant. The test yielded

$$U = 56303886, \quad p < 0.001,$$

indicating that the distributions of reviews per month differ significantly between the two host groups. The test results are summarized in Table, which reports median values, the U statistic, and the p value.

Because engagement is influenced by availability, an availability adjusted engagement metric was created. This variable normalizes review frequency by availability through the expression

$$\text{engagement_adj} = \frac{\text{reviews_per_month}}{\max(\text{availability_365}/365, 0.05)}$$

A boxplot comparing the adjusted metric across host types showed a smaller difference between Single and Multi hosts, suggesting that availability could explain part of the engagement gap.

A second Mann Whitney U test confirmed this pattern. For the adjusted engagement metric, the test produced

$$U = 60898669, \quad p < 0.001.$$

The statistical significance remained, but the magnitude of the difference was smaller, indicating that availability contributes to observed engagement differences.

To examine these relationships while controlling for listing characteristics, two regression models were constructed. Both used log base ten transformations to improve interpretability and stabilize variance. The first model used

$$\text{log10_rpm} = \log_{10}(1 + \text{reviews_per_month})$$

and the second used

$$\text{log10_engagement_adj} = \log_{10}(1 + \text{engagement_adj})$$

Key regression findings (Model 1: log10 reviews per month):

- Coefficient for **Single hosts**: **0.0136**, **p < 0.001**
- Percentage effect:

$$(10^{0.0136} - 1) \times 100 \approx 1.4\%$$

Key regression findings (Model 2: availability adjusted engagement):

- Coefficient for **Single hosts**: **-0.0055**, **p = 0.327**
- Percentage effect:

$$(10^{-0.0055} - 1) \times 100 \approx -0.5\%$$

- Effect is **not statistically significant**

Model performance metrics indicated that both models explained a modest proportion of variance in engagement, which is expected given the behavioral complexity of guest reviewing patterns (Figure 9.13 and Table 9.6, which you will insert).

Model	Dependent Variable	Host Type Effect	p-value	Interpretation
Model 1	log ₁₀ (Reviews per Month)	+0.0136 → +1.4%	< 0.001	Single hosts receive ~1.4% more reviews per month than multi hosts.
Model 2	log ₁₀ (Availability-Adjusted Engagement)	-0.0055 → -0.5%	0.327	Once availability is controlled, engagement difference disappears.

Key numerical results:

- Median reviews per month
 - **Single hosts: 0.81**
 - **Multi hosts: 0.54**
- Mann Whitney U statistic (raw engagement): **56,303,886**, **p < 0.001**
- Mann Whitney U statistic (adjusted engagement): **60,898,669**, **p < 0.001**
- Regression effect for Single hosts on raw engagement: **+1.4 percent**
- Regression effect for Single hosts on adjusted engagement: **-0.5 percent (not significant)**

Interpretation

These combined findings show that Single hosts receive more guest reviews per month than Multi hosts on average. However, once engagement is normalized for availability and key listing features such as location, room type, property size, rating, and price, the effect of host strategy is no longer statistically significant. This indicates that the higher engagement observed for Single hosts is primarily driven by factors such as availability patterns, pricing, or listing characteristics rather than host strategy alone.

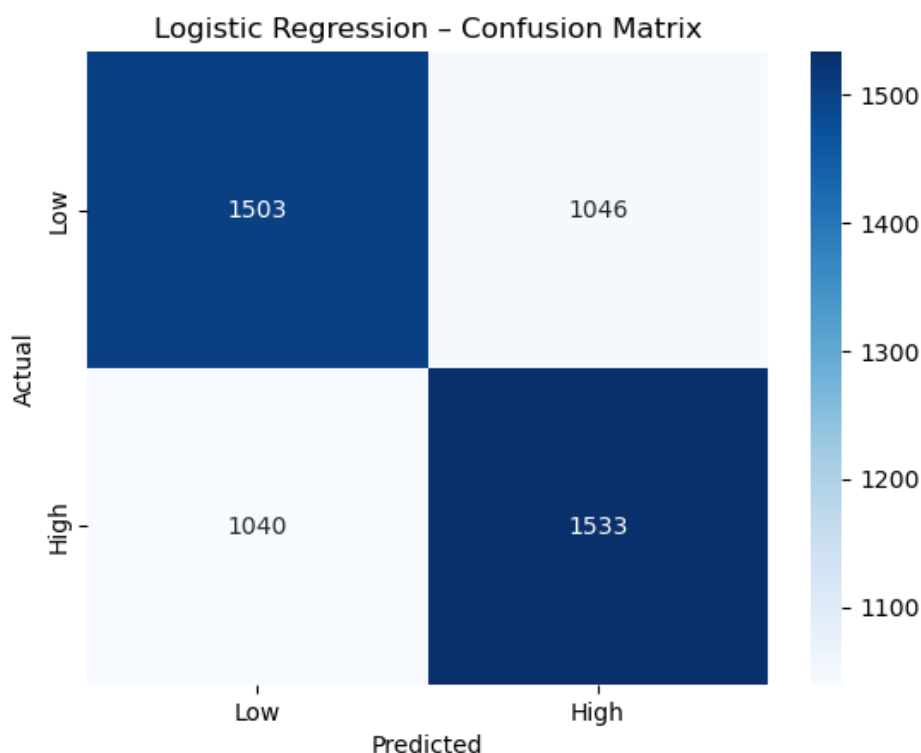
In summary, host strategy appears to influence raw engagement but does not exert an independent effect once availability and listing attributes are taken into account. This result distinguishes between operational effects and true engagement effects, providing a nuanced understanding of how host behavior relates to guest reviewing patterns.

- **RQ5: Can listings be classified into high demand versus low demand segments based on review activity?**

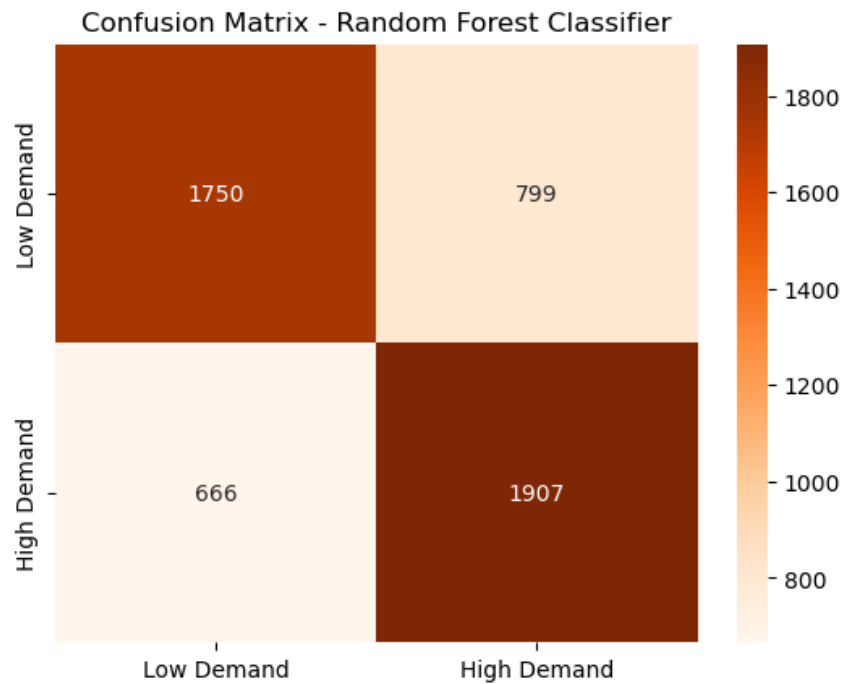
This research question examines whether Airbnb listings can be classified into high demand and low demand groups using review activity as a measure of guest engagement. Since reviews per month were highly skewed and demonstrated substantial variation across listings, supervised learning methods were applied to determine how effectively listing characteristics predict demand status.

High demand listings were defined as those with reviews per month at or above the median value, while low demand listings were defined as those below the median. Three classification models were evaluated: logistic regression, random forest, and histogram based gradient boosting. Each model incorporated predictors such as log price, room type, borough, beds, bedrooms, bathrooms, rating, availability, and host type. Model performance was assessed using accuracy, ROC AUC, and confusion matrices.

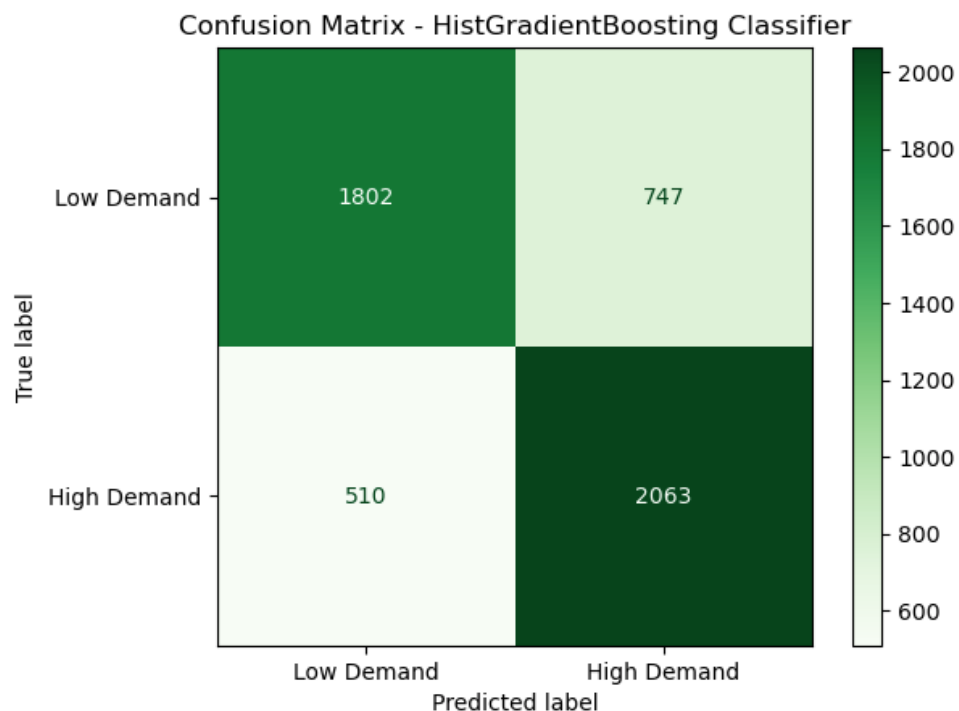
The logistic regression model served as the baseline classifier. It achieved an accuracy of **0.593** and a ROC AUC of **0.634**, indicating modest predictive ability. The confusion matrix for this model showed that predictions were distributed relatively evenly across true positives and true negatives, but the model misclassified a large portion of both categories. This confirmed that a linear decision boundary was insufficient for capturing the nonlinear relationships underlying review based demand.



The random forest classifier produced noticeably stronger performance. It achieved an accuracy of **0.714** and a ROC AUC of **0.786**. The confusion matrix showed substantial improvement in the correct classification of high demand listings, with fewer false negatives compared to logistic regression. Feature importance output indicated that review activity interacts in complex ways with room type, rating, availability, and pricing, which the random forest captured effectively.



The histogram based gradient boosting classifier delivered the strongest predictive accuracy among all models. It achieved an accuracy of **0.755** and a ROC AUC of **0.825**, representing the best classification quality in the study. The confusion matrix showed **2063 correct high demand predictions** and **1802 correct low demand predictions**, with substantially reduced errors relative to the baseline model. This confirmed that the gradient boosting classifier learned nonlinear and interaction based patterns that were not captured by simpler methods.



Model	Accuracy	ROC-AUC	Interpretation
Logistic Regression	0.593	0.634	Linear model captures limited relationships; demand is complex and nonlinear.
Random Forest Classifier	0.714	0.786	Captures nonlinear interactions; strong improvement over baseline.
HistGradientBoosting	0.755	0.825	Best overall performance; models complex interactions between pricing, rating & availability.

Key classification performance results:

- Logistic Regression
 - Accuracy: **0.593**
 - ROC AUC: **0.634**
- Random Forest Classifier
 - Accuracy: **0.714**
 - ROC AUC: **0.786**
- Histogram Based Gradient Boosting Classifier
 - Accuracy: **0.755**
 - ROC AUC: **0.825**

Interpretation

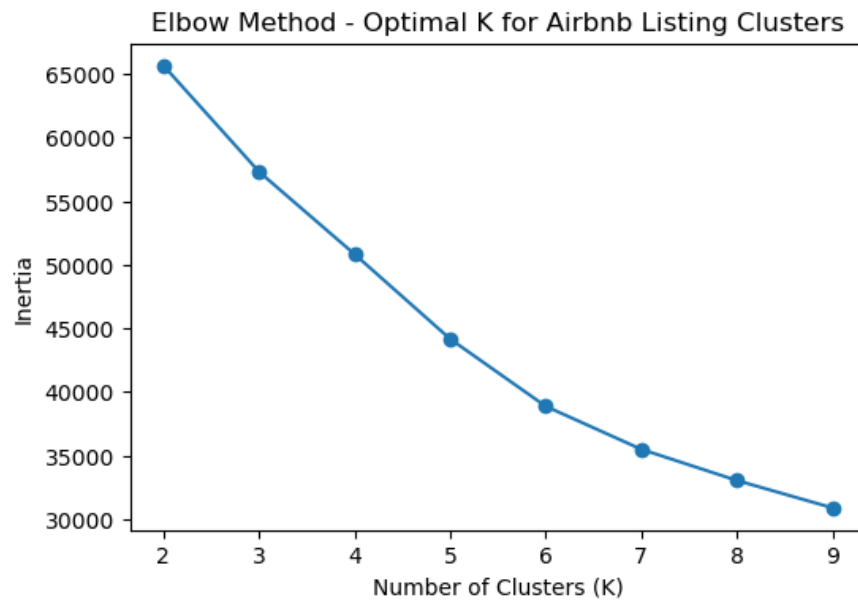
The combined results demonstrate that Airbnb listings can be classified into high and low demand segments; however, this is only possible when nonlinear models are used. Logistic regression was unable to capture the complexity of review based demand patterns, while the random forest and histogram based gradient boosting classifiers produced strong and meaningful segmentation. The gradient boosting model, in particular, identified high demand listings with high accuracy and revealed that listing characteristics such as rating, room type, availability, and price jointly influence review activity.

In summary, demand segmentation is present in the New York City Airbnb market, and machine learning models, especially nonlinear ensemble methods, provide effective tools for distinguishing between listings with high and low levels of guest engagement.

• RQ6: What clusters of listings exist when considering price, host behavior, reviews, and ratings?

This research question identifies natural market segments in the New York City Airbnb dataset using an unsupervised learning approach. Clustering was performed using K means on standardized variables that capture essential listing characteristics. The selected features included log transformed price, rating, reviews per month, availability, and a binary indicator for host type. These variables represent pricing, quality, demand, availability, and host strategy, which collectively shape listing performance.

The optimal number of clusters was determined using the Elbow Method. The plot of within cluster sum of squares against the number of clusters showed a clear inflection point at **K = 4**, indicating that four clusters provided a meaningful balance between granularity and interpretability. K means clustering with four clusters was therefore used for segmentation.



The resulting clusters are summarized in Table, which reports the mean values of log price, rating, reviews per month, availability, and host type for each cluster. These cluster centroids provide a profile of the typical listing within each group. The clusters also had distinct sizes, with Cluster 0 containing **7345** listings, Cluster 1 containing **4501** listings, Cluster 2 containing **1297** listings, and Cluster 3 containing **3638** listings. A PCA projection visually confirmed that the four clusters are well separated in reduced dimensional space.

	log_price	rating	reviews_per_month	availability_365	host_type_encoded	count
cluster						
0	5.00	4.82	1.51	182.92	1.00	7345
1	4.82	4.78	1.12	320.75	0.00	4501
2	4.62	3.99	0.62	237.85	0.23	1297
3	4.69	4.76	1.87	86.32	0.00	3638

Cluster Profiles:

Cluster 0: Mid priced single host listings with strong engagement

- Average log price: **5.00**
 - Average rating: **4.82**
 - Reviews per month: **1.51**
 - Availability: **182.9 days**
 - Host type encoded: **1.00** (predominantly Single hosts)
- This cluster represents individual hosts offering mid range listings with strong guest engagement and above average ratings.

Cluster 1: High availability, lower priced multi host listings

- Average log price: **4.82**
- Average rating: **4.78**
- Reviews per month: **1.12**
- Availability: **320.8 days**
- Host type encoded: **0.00** (Multi hosts)

This segment consists of properties managed by Multi hosts with broad availability and competitive pricing aimed at maximizing occupancy.

Cluster 2: Low quality and low engagement listings

- Average log price: **4.62**
- Average rating: **3.99** (lowest among all clusters)
- Reviews per month: **0.62**
- Availability: **237.9 days**
- Host type encoded: **0.23**

This cluster captures underperforming listings that receive fewer reviews and have the lowest ratings, suggesting quality issues or poor market fit.

Cluster 3: Budget high demand listings with limited availability

- Average log price: **4.69**
- Average rating: **4.76**
- Reviews per month: **1.87** (highest engagement)
- Availability: **86.3 days** (lowest availability)
- Host type encoded: **0.00** (Multi hosts)

This group represents budget friendly listings with high booking turnover, reflected by low availability and strong review activity.

Interpretation

The four clusters reveal distinct market segments within the New York City Airbnb landscape. Cluster 0 captures mid priced listings operated by Single hosts who appear to offer higher quality and receive strong engagement. Cluster 1 reflects volume driven strategies typical of Multi hosts. Cluster 2 consists of low quality listings with limited demand. Cluster 3 represents a unique high turnover, budget oriented segment with the strongest review frequency across the dataset.

These findings highlight the diversity of host strategies and listing types within the market. The clusters demonstrate that price, host behavior, quality ratings, and review activity jointly shape distinct groups of listings, providing valuable insight into the market structure.

⇒ Conclusion

This study examined pricing dynamics, host behavior, demand segmentation, and market structure in the New York City Airbnb market using a combination of descriptive statistics, inferential testing, regression modeling, classification methods, and clustering. The analysis used a dataset containing more than twenty thousand listings and incorporated both structural listing characteristics and behavioral indicators to understand the drivers of price and guest engagement.

The first set of findings demonstrated that nightly prices differ substantially across boroughs and room types. Manhattan and Brooklyn consistently occupied the highest price tiers, while Queens, the Bronx, and Staten Island represented more affordable markets. Entire homes and hotel rooms were priced far above private and shared rooms. Statistical tests confirmed that these differences were highly significant and reflect meaningful economic segmentation across the city.

Regression modeling showed that price is driven primarily by supply side attributes. Room type and borough were the strongest predictors, with property size variables such as beds, bedrooms, and bathrooms contributing additional explanatory power. Ratings and host strategy had smaller but still meaningful effects. The histogram based gradient boosting model provided improved predictive accuracy compared to linear regression, demonstrating that price formation involves nonlinear interactions among listing characteristics.

The role of host strategy was examined through comparisons between Single and Multi hosts. Single hosts consistently charged higher prices than Multi hosts, and statistical tests confirmed that this difference was significant. Regression results showed that Single hosts priced their listings approximately seventeen percent higher than Multi hosts after controlling for location, room type, listing size, and rating. These findings suggest that host strategy influences pricing behavior, potentially reflecting differences in listing quality, personalization, or operational focus.

Guest engagement, measured through review activity, was also found to vary across host types. Single hosts showed slightly higher review activity in raw form, but once engagement was adjusted for availability and other factors, this difference was not statistically significant. This indicates that differences in engagement are influenced by operational variables rather than host strategy alone.

The classification models used to distinguish high demand from low demand listings showed that demand segmentation exists within the dataset. Logistic regression provided only modest accuracy, while nonlinear models such as random forest and histogram based gradient boosting achieved significantly stronger performance. These models showed that review based demand is influenced by a combination of pricing, rating, availability, room type, and host characteristics.

Clustering analysis revealed four distinct market segments based on price, host strategy, review activity, ratings, and availability. These clusters represented mid priced single host listings with strong engagement, high availability multi host listings, low quality low demand listings, and budget oriented high turnover listings. These segments provided insight into how different combinations of characteristics shape market behavior in New York City.

Overall, the results show that the Airbnb market in New York City is shaped by a combination of supply side characteristics, host strategies, geographic patterns, and guest engagement behaviors. Price is primarily determined by accommodation type and location, while demand patterns are best captured through nonlinear models that incorporate multiple listing attributes. The findings of this study offer valuable implications for hosts who seek to optimize pricing and listing management, as well as for platform stakeholders interested in understanding the structure and dynamics of urban short term rental markets.